	minimum separation = 2s = 0.0682 m	A1
3(a)	Equal and opposite forces act on the two trolleys (Newton's Third Law) at every instant.	M1
	By Newton's 2nd law, rate of change of momentum will be the same for the two trolleys.	M1
	Since they act for the same time interval and have same force, the impulses (change in momentum) are equal in magnitude (and opposite in direction).	A1

(b)	No resultant external horizontal force acts on the system / Both trolleys experience the same change in momentum	M1
	Therefore the total momentum remains zero and hence the centre of gravity remains at rest.	A1
	By Newton's first law, the CG hence remains at rest.	
(c)	Applies conservation of momentum $$M_1 v_1 = M_2 v_2.$$	M1
	Uses $$KE = \frac{1}{2} m v^2$$ **and** substitutes/eliminates one velocity correctly. Alternatively, accept KE = p^2 / (2m) used in conjunction with the first equation.	M1
	Obtains $$\frac{KE_A}{KE_B} = \frac{M_2}{M_1}$$	A1
4(a)(i)	The graph is a **straight** line that passes through the **origin**. This indicates that the acceleration is **directly proportional** to the displacement, which is	B1